SPECIAL SESSION · 7 SLIDES · BANKING & FINANCE

Loop Engineering

You already have the loop. Every agent is one. The discipline is what you wrap around it: where the state lives, what decides the work is finished, and on whose clock the whole thing repeats.

Loop Engineering | State Outside the Model | The Verifier | create_agent | Budgets



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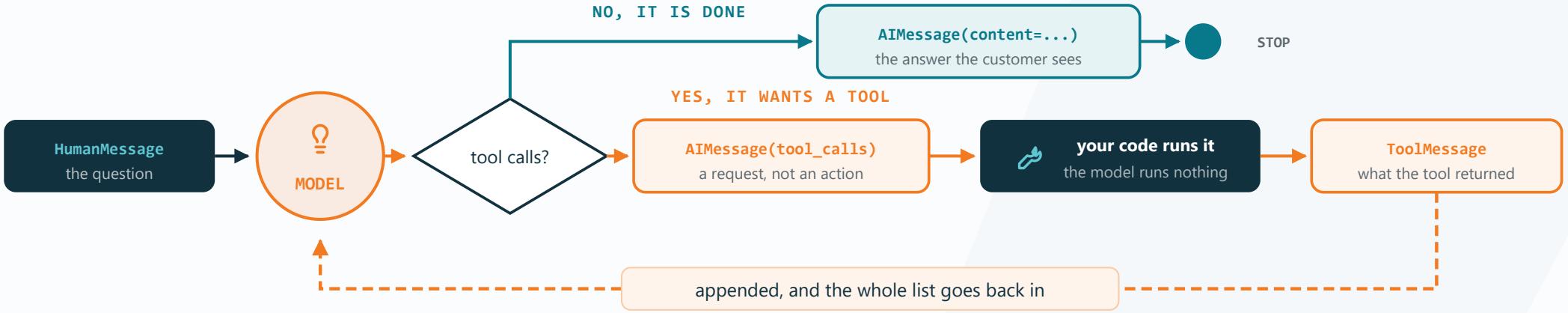
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You Have Seen This Before

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LOOKS LIKE the agent loop. ReAct. Tool calling.

It is. Loop engineering is what wraps it.



The model never executes

It writes a request and waits. Every guard you want lives in the gap you own.

Nothing here counts

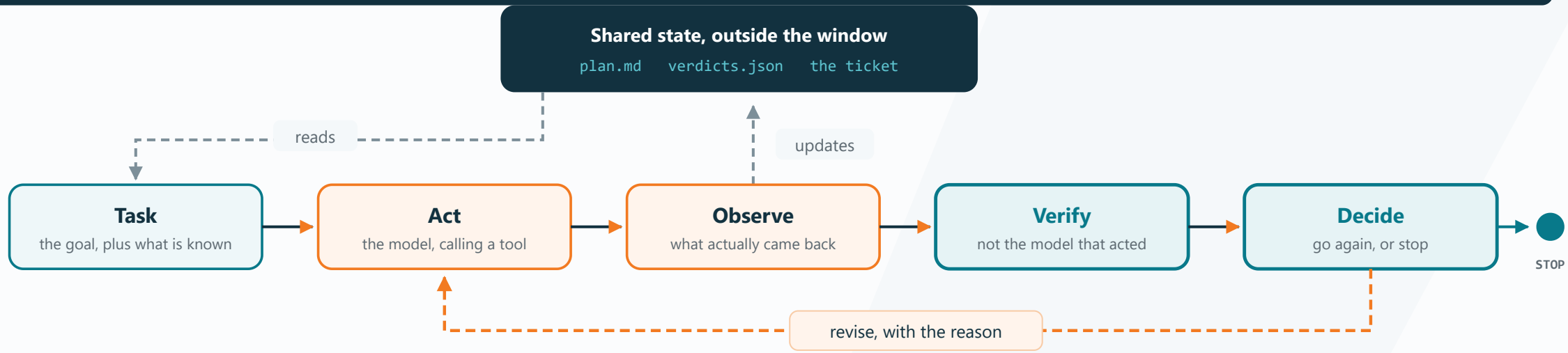
There is no natural limit in the picture. If you want one, you have to add it.

Nothing here remembers

Everything the run knows is in that list, and the list dies with the run.

Five Steps, and Two of Them Are Not the Model

LOOKS LIKE a while loop with extra steps. The state and the check sit outside the model.



AND IT RUNS AT THREE SPEEDS AT ONCE

INNER · MINUTES

The agent acts, checks, and fixes.

The check is a test that passes or fails.

MIDDLE · HOURS

You read the output and steer it.

The check is your judgement.

OUTER · WEEKS

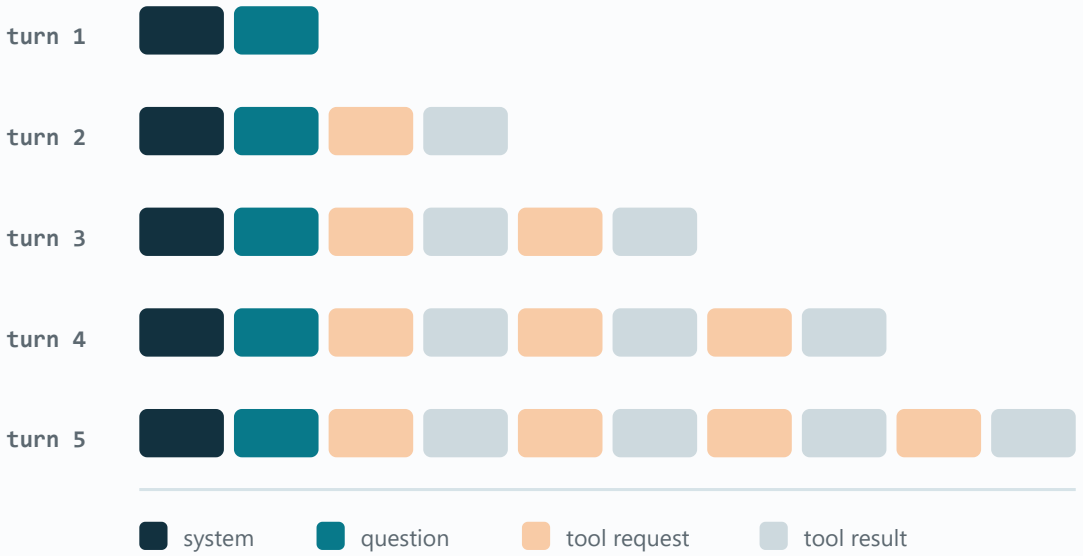
Customers use it and react.

The check is the number that moved.

If the Memory Is the Transcript, It Is a Chat

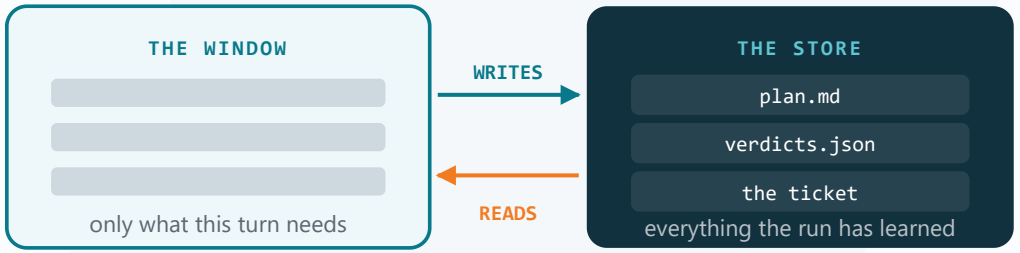
LOOKS LIKE a database, or a scratch file. That is exactly what it is.

THE SYMPTOM · WHAT GOES UP THE WIRE



Only the last two blocks are new. The rest is paid for again,
and the useful signal in the pile keeps thinning. That is context rot.

THE FIX · WHERE THE STATE SHOULD LIVE



THE THREE MOVES

- ✓ **Compaction** summarise the run so far, continue from the summary
- ✓ **Offloading** push big outputs to a file, keep only the slice you need
- ✓ **Sub-agents** hand a messy subtask out, only its clean result returns

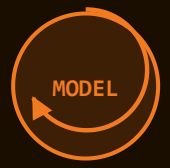
Turn nine does not need turn one in the window.
It needs turn one on disk.

The Maker Cannot Grade Itself

LOOKS LIKE code review, or an LLM as a judge.

The verdict has to be a value, not prose.

SELF-GRADING · THE SHAPE EVERYONE BUILDS FIRST

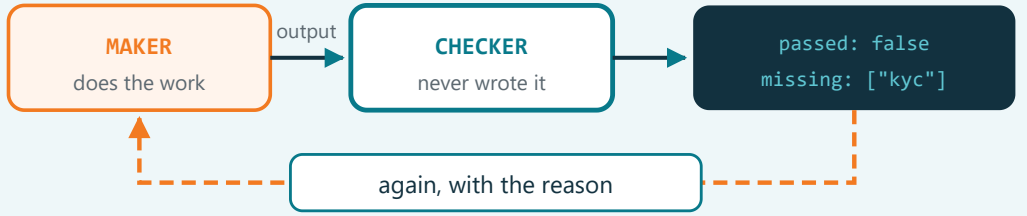


writes it, then reviews it

```
"Looks correct to me."
```

The same weights that produced the answer decide whether the answer is good. It doubles the calls **and moves the verdict least.**

A SEPARATE CHECK · WHAT ACTUALLY CLOSES IT



WHAT MAKES A CHECK WORTH HAVING

1 · A VALUE, NOT PROSE

```
{"passed": false,
"missing": ["kyc"]}
```

A test suite, a schema, a type checker.

Something your code can branch on.

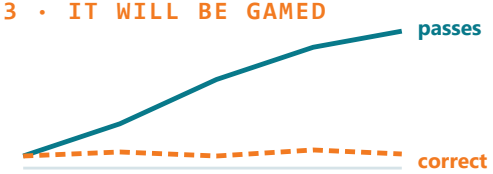
2 · NOT THE MAKER



A small cheap model beats a large one **grading itself.**

Different prompt, different context, no memory of writing it.

3 · IT WILL BE GAMED



Loops learn to satisfy the checker. Re-read what passed.

Four Ways It Fails, and What Catches Each

PYTHON · LANGCHAIN 1.X · CALL LIMITS

```python
from langchain.agents import create_agent
from langchain.agents.middleware import ModelCallLimitMiddleware
from langgraph.checkpoint.memory import InMemorySaver

agent = create_agent(
    model=model,
    tools=[get_statement, calculate_emi],
    checkpointer=InMemorySaver(),      # thread_limit needs one
    middleware=[ModelCallLimitMiddleware(
        run_limit=5,                # this one invocation
        thread_limit=20,            # the whole conversation
        exit_behavior="end",         # stop politely, do not raise
    )],
)

# a second, independent ceiling, one level down in the runtime
agent.invoke(question, config={"recursion_limit": 12})

# The budget cap and the no-progress check are not in the
# framework. You write those. No-progress is simply: did
# this turn repeat the previous call with the same arguments?

# And none of them know whether the work is finished. They
# only know it has run long enough that something is wrong.
```

FOUR FAILURES, FOUR BRAKES

1 · IT NEVER STOPS

The tell: finance notices before engineering does.

CAUGHT BY — max iterations, and a budget cap

2 · IT STOPS TOO EARLY

The tell: nobody notices at all.

CAUGHT BY — the completion check, and nothing else

3 · IT REPEATS ITSELF

The tell: your tool returned a shrug, not a value.

CAUGHT BY — no-progress detection

4 · IT LEARNS TO PASS THE CHECK

The tell: the rubric passed, the customer did not.

NO BRAKE — you have to re-read what passed

Four Things to Keep

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State goes outside

If the run's memory is the message list, it dies with the run and it costs more every turn until it does. A file on disk fixes both at once.

Something else grades it

A model asked whether it is happy with its own work will say yes. The verdict has to be a value your code can branch on, not a paragraph.

The clock is yours

Minutes, hours, weeks. Each layer needs its own check and its own audience. A loop with no cadence is just a person pressing enter repeatedly.

Budgets are the backstop

Set run_limit and recursion_limit on day one, and then log which one fires. If it is not the check, the work was never finished.

Can you check the answer without asking the model?

YES, SO IT CAN CLOSE ITSELF
Wire the check in and let the loop run on its own clock.

NO, SO YOU ARE THE CHECK
Keep it short, keep it watched, and cap it hard.